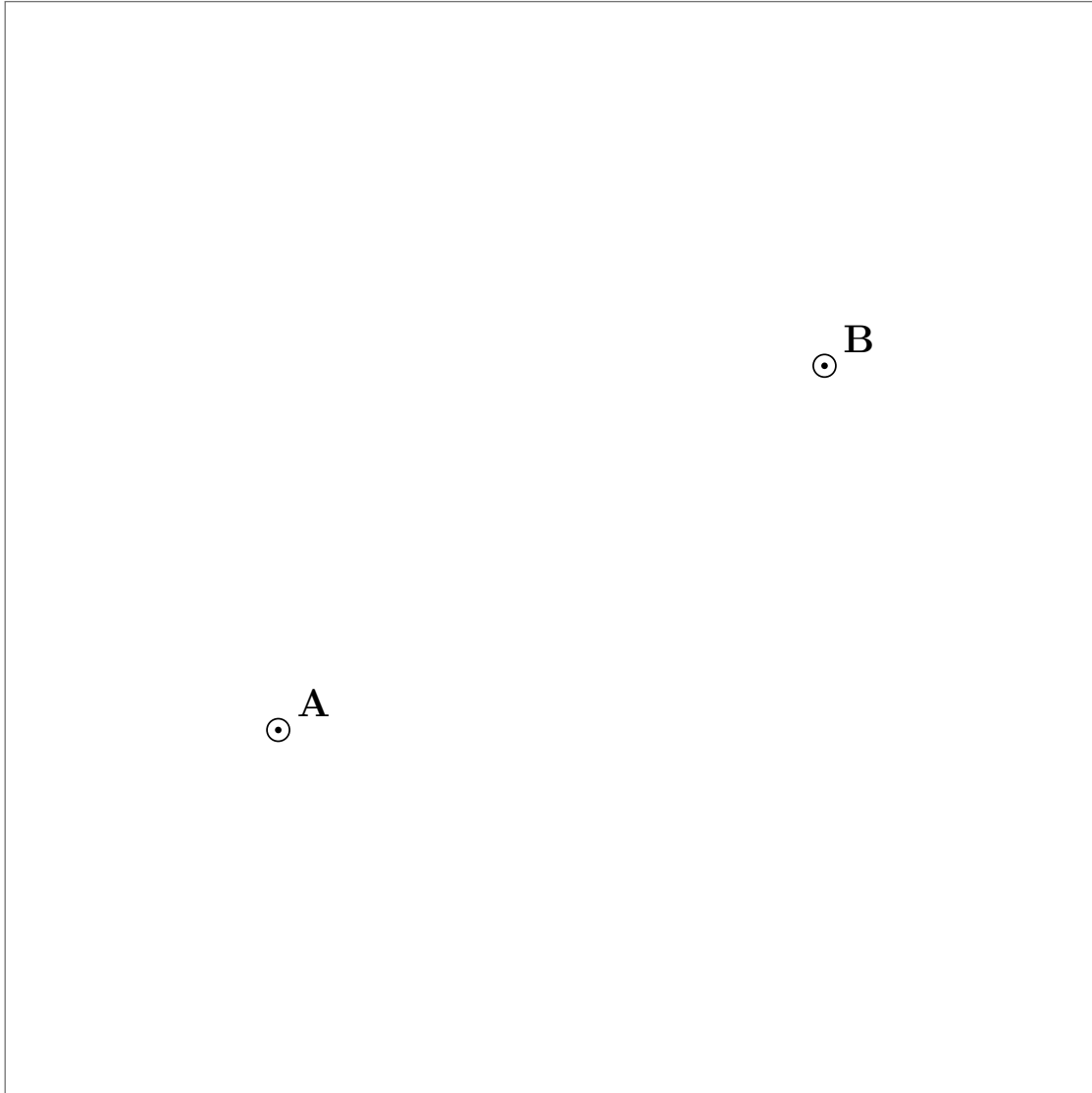
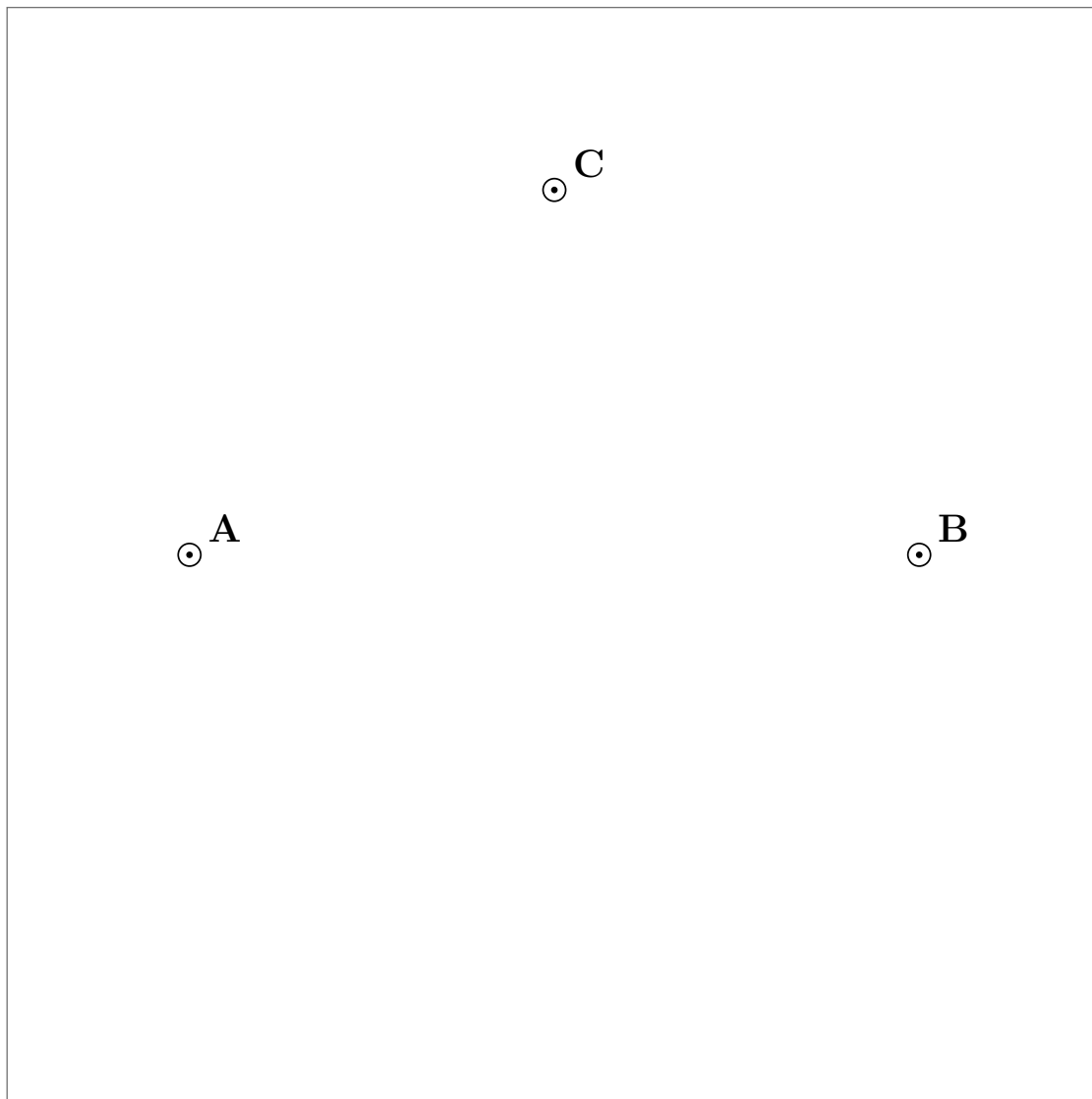


Keep the dots fixed and in different places. Use straight-line distance to their centers. A dot's region includes every place where it is nearest, including shared ties. The frame shows part of a larger plane.

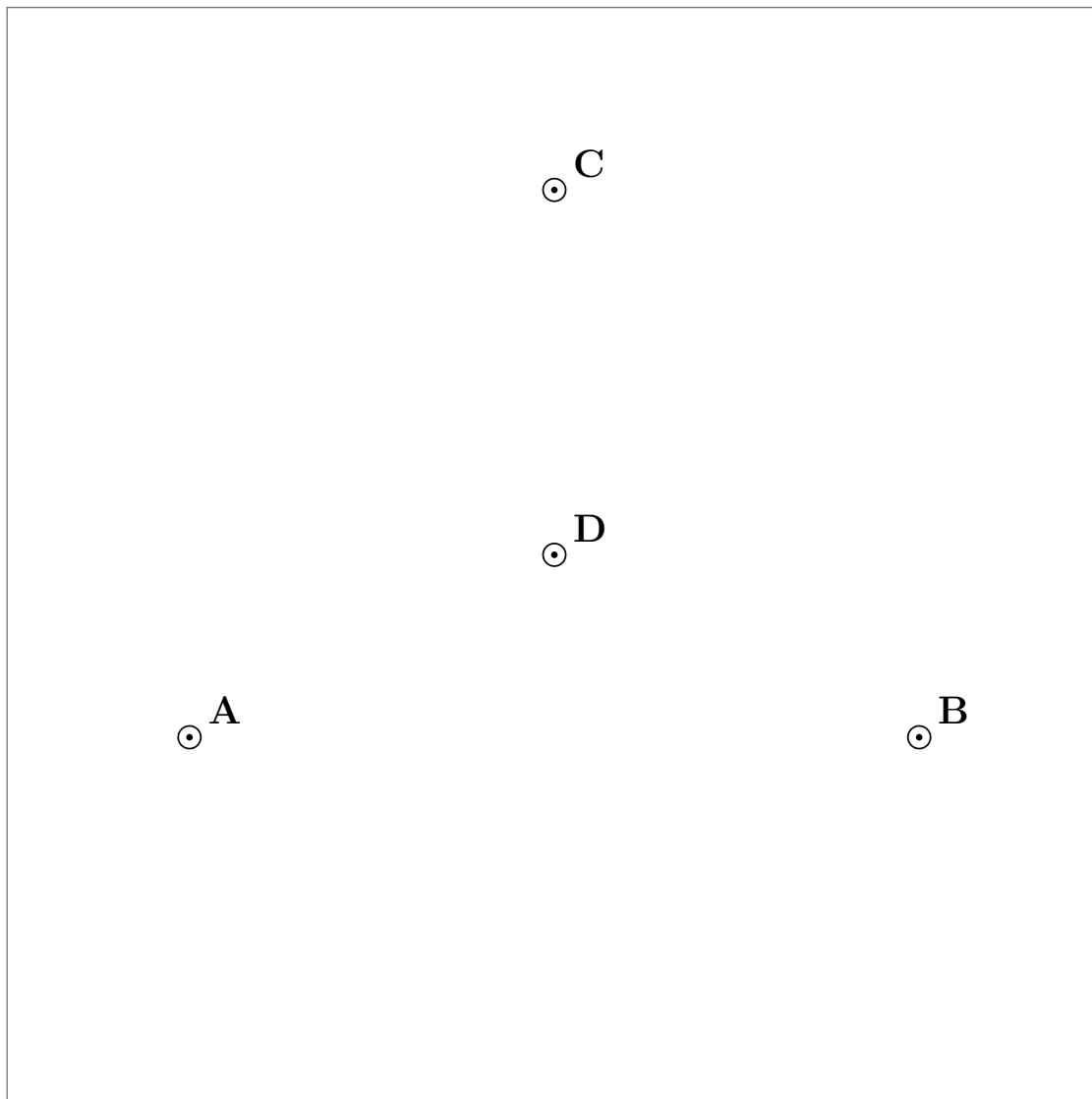
Problem 1: Draw the two nearest-dot regions. Explain why your boundary works for every point on the sheet.



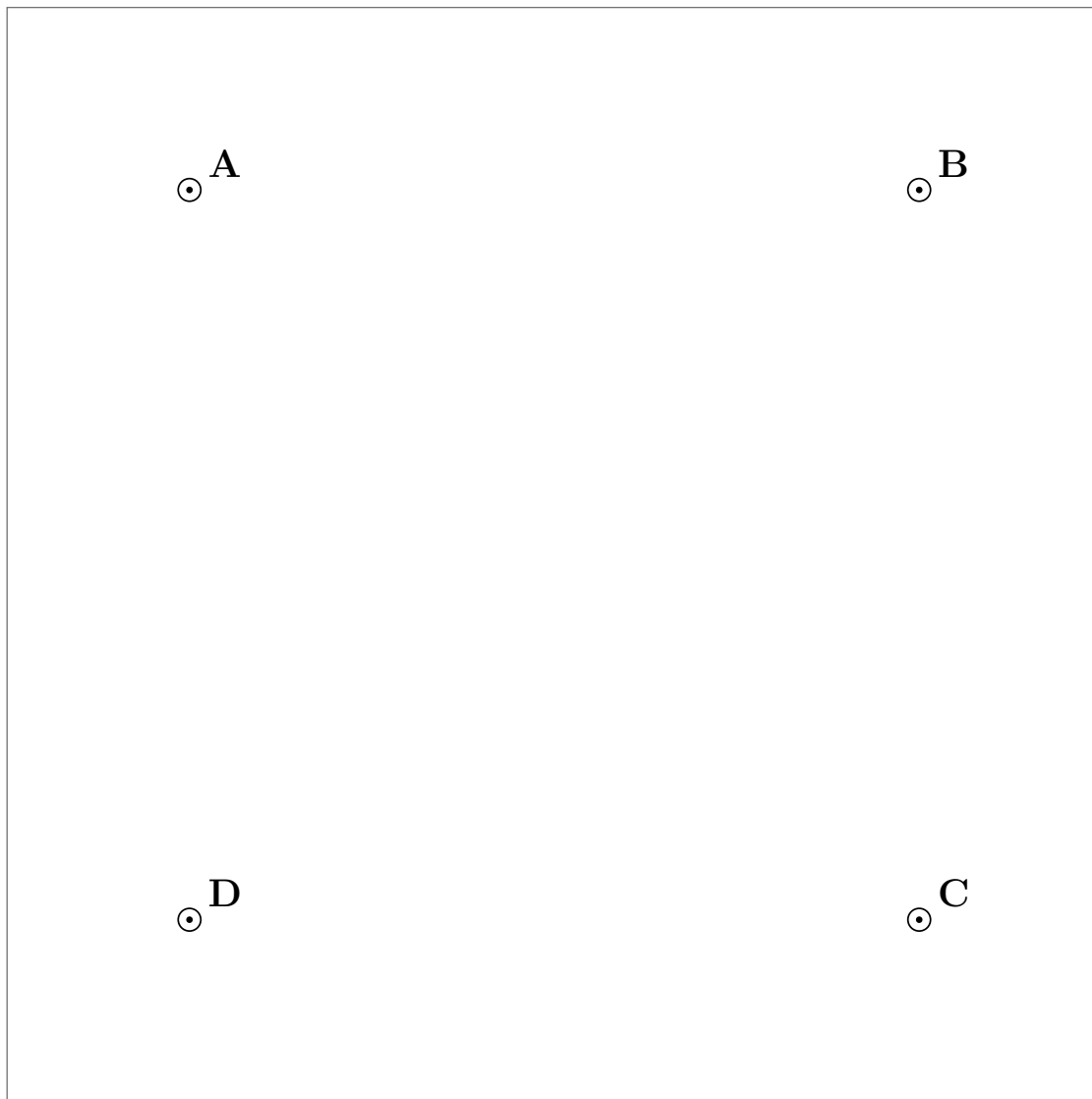
Problem 2: Draw the three regions, including all the places shared by nearest dots.



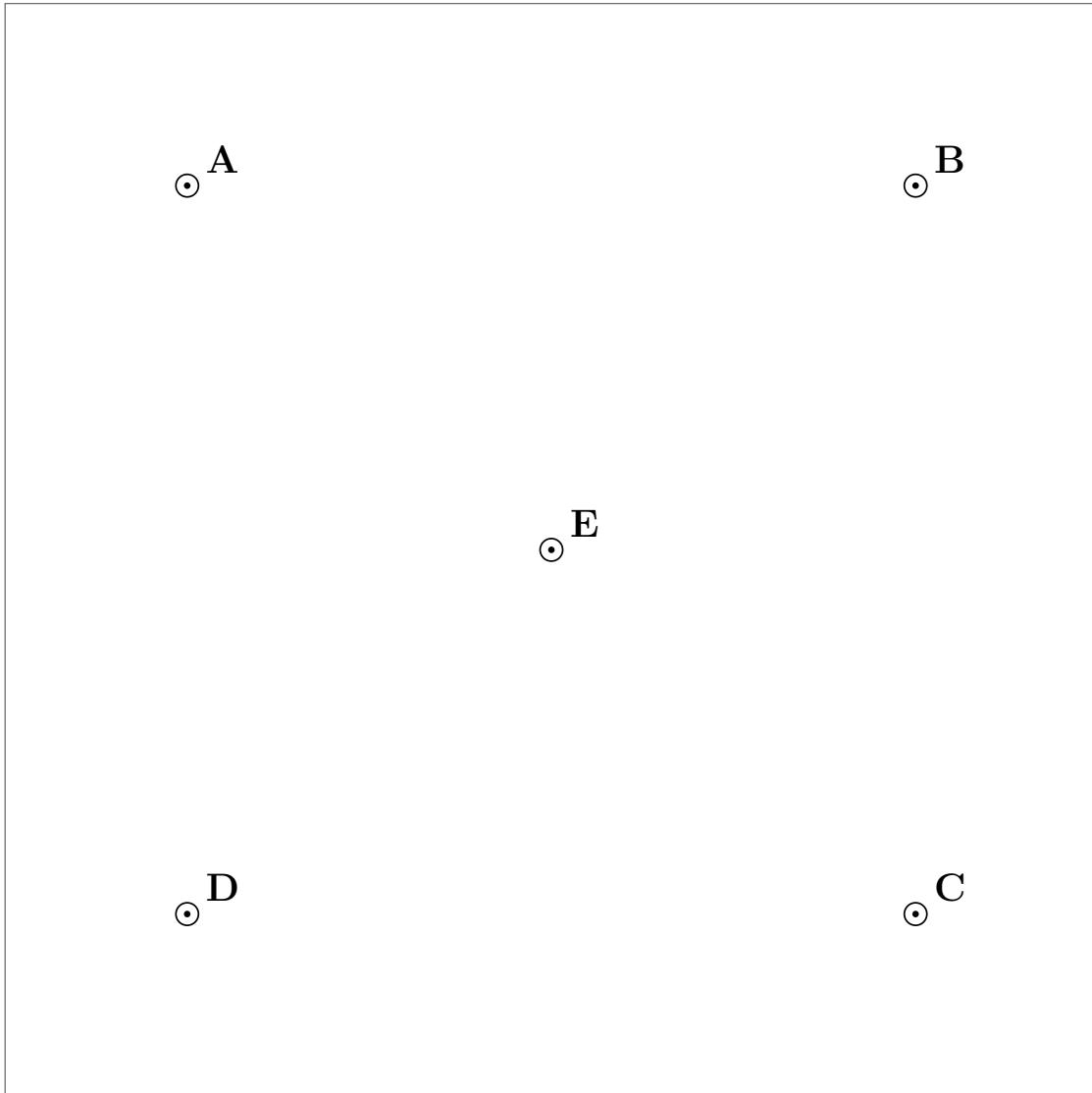
Problem 3: Draw D's whole region. Can a straight segment joining two points of that region ever leave it? Explain why your answer holds for every pair of points.



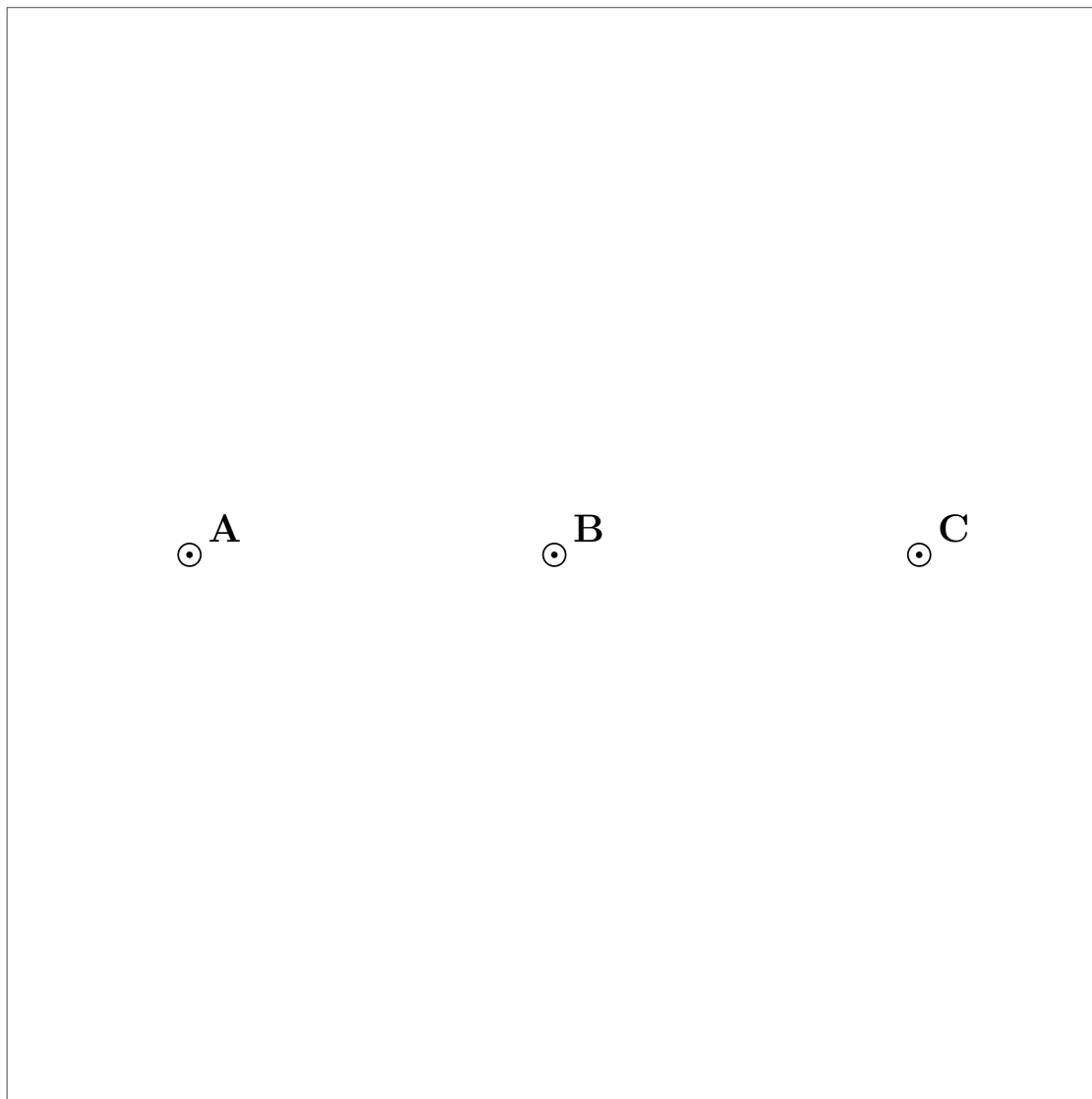
Problem 4: Draw the four regions. Find every place shared by three or more nearest dots.



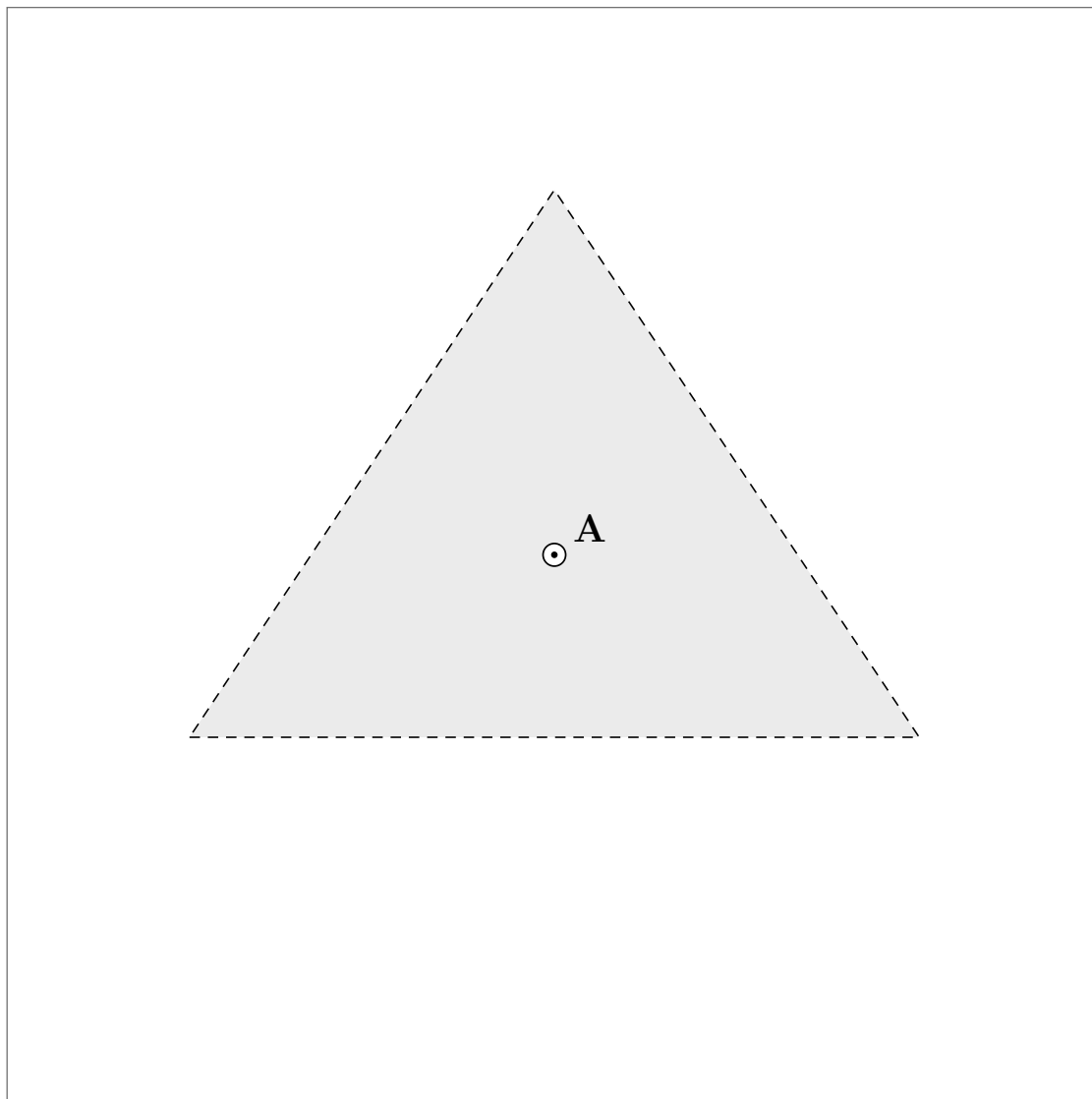
Problem 5: E is added to the dots in Problem 4. Draw all five regions and mark the old boundaries that disappear. Can adding a new dot ever give an old dot more places? Explain.



Problem 6: Place as few new dots as you can so B's entire region fits inside the frame. Draw that region and explain why fewer new dots cannot work.



Problem 7: Place three new dots so A's whole region is exactly the shaded triangle.



Problem 8: For R and S separately, decide whether new dots can take that point away from A while P and Q still belong to A. Draw a placement when it is possible; explain when it is impossible.

